NAMRATA DUTT

PhD Candidate, Agricultural and Biological Engineering | University of Florida

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EDUCATION

University of Florida, *Gulf Coast Research and Education Center, Wimauma, Florida* Jan 2023-present

PhD Candidate, Agricultural and Biological Engineering | Advisor: Daeun Choi; GPA: 3.72/4.00

University of Florida, *Gainesville, Florida* Aug 2021-Dec 2022

MS, Computer Science | GPA: 3.66/4.00

Banasthali Vidyapith, *Rajasthan, India* Aug 2017-May 2021

BTech, Computer Science | GPA: 7.33/10.00; final semester coursework at the University of Florida, Jan-May 2021 (GPA: 3.83/4.00)

RESEARCH FOCUS

Computer vision and machine learning researcher using color and depth (RGB-D) imaging to detect, track, and assess the maturity of individual crops, supporting robotic harvesting and crop monitoring.

RESEARCH EXPERIENCE

Graduate Research Assistant, *University of Florida, Gulf Coast Research and Education Center* Jan 2023-present

- Developed RGB-D computer vision methods for white button mushroom detection, instance segmentation, localization, and temporal tracking to support selective robotic harvesting.
- Developed RD-GuideNet, a depth-guided detection, segmentation, and tracking framework for white button mushrooms, achieving an F1-score of 0.93 for segmentation and 92.7% tracking consistency.
- Built maturity assessment methods by calculating mushroom geometry, depth, and temporal growth features, resulting in an MAE of 5.77, including ongoing work on crowding effects and time-to-harvest prediction.
- Designed and curated longitudinal RGB-D imagery and contributed to a published time-series dataset for mushroom instance segmentation.

Graduate Research Assistant, *University of Florida, Agricultural and Biological Engineering* Aug-Dec 2022

Supervisor: Aditya Singh

- Developed deep learning approaches for land-cover change detection and analysis of UAV hyperspectral imagery and LiDAR data for crop health assessment.

PUBLICATIONS AND RESEARCH OUTPUTS

JOURNAL ARTICLES

Dutt, N., Choi, D., Ampatzidis, Y., Lee, W. S., Koppal, S. J., & Wang, X. (2026). RD-GuideNet: A depth-guided framework for robust detection, segmentation, and temporal tracking of white button mushrooms. Sensors, 26(6), 1935. **Editor's Choice.** https://doi.org/10.3390/s26061935

Dutt, N., & Choi, D. (2026). An RGB-D time-series dataset of white button mushroom growth for instance segmentation. Data in Brief, 68, 113085. https://doi.org/10.1016/j.dib.2026.113085

EXTENSION ARTICLE

Dutt, N., & Choi, D. (2025). How AI can analyze images: An introductory case study using mushroom detection. EDIS, 2025(6). https://doi.org/10.32473/edis-ae614-2025

CONFERENCE PAPER

Dutt, N., & Choi, D. (2024). A computer vision system for mushroom detection and maturity estimation using depth images. 2024 ASABE Annual International Meeting. American Society of Agricultural and Biological Engineers.

PUBLISHED DATASET

Dutt, N., Choi, D., & Pecchia, J. (2026). RGB-D time-series dataset of white button mushroom growth for instance segmentation (Version 1) [Data set]. Mendeley Data. https://doi.org/10.17632/8n6nr43gk3.1

PRESENTATIONS

ORAL PRESENTATIONS

Dutt, N., & Choi, D. (2026). From prediction to harvest: AI-driven mushroom maturity estimation and harvest scheduling [Lightning talk]. UF AI Days, University of Florida, Gainesville, Florida.

Dutt, N., & Choi, D. (2026). A two-stage framework for mushroom maturity estimation and time-to-harvest prediction using 3D depth data. ASABE Annual International Meeting, Indianapolis, Indiana.

Dutt, N., & Choi, D. (2025). 3D point cloud reconstruction for automated mushroom segmentation and growth monitoring. ASABE Annual International Meeting, Toronto, Canada.

Dutt, N., & Choi, D. (2024). A computer vision system for mushroom detection and maturity estimation using depth images. ASABE Annual International Meeting, Anaheim, California.

POSTER PRESENTATIONS

Dutt, N., & Choi, D. (2026). A two-stage framework for mushroom maturity estimation and time-to-harvest prediction using 3D depth data. South Florida Student and Post-Doc Research Symposium, Homestead, Florida.

Dutt, N., & Choi, D. (2025). An integrated approach for RGB-D acquisition, precision segmentation, and mushroom growth pattern analysis. South Florida Student and Post-Doc Research Symposium, Belle Glade, Florida.

Dutt, N., & Choi, D. (2025). Precision mushroom farming: Detection and maturity estimation with depth imaging. University of Florida Graduate Student Research Day, Gainesville, Florida.

Dutt, N., & Choi, D. (2024). Automated mushroom detection and maturity estimation using depth images. University of Florida Inter-REC Poster and Flash Talk Competitions, Wimauma, Florida.

EXTENSION AND PUBLIC ENGAGEMENT

Presenter. "Can a Computer Tell a Mushroom Is Ready to Pick? When AI Meets Agriculture: Smart Tools, Smarter Farming." Expanding Agriculture: Workforce Program, College of Central Florida, Lecanto, Florida, Sept 2026.

Presenter. Mushroom detection and AI demonstration, UF/IFAS Gulf Coast Research and Education Center Centennial Celebration and AI Center Groundbreaking, Wimauma, Florida, Nov 2025.

Presenter. Mushroom detection and AI demonstration for growers and industry representatives, Florida Strawberry Growers Association Tailgate, Dover, Florida, Nov 2025.

Presenter. Depth sensor and mushroom detection demonstration. UF/IFAS Range Cattle Research and Education Center Youth Field Day, Ona, Florida, June 2025.

Presenter. “Linear-Rail Imaging for Mushroom Detection and Maturity Estimation.” Research tour for Professor Elizabeth Sklar, UF/IFAS Gulf Coast Research and Education Center (GCREC), November 10, 2025.

Presenter. “Automated Mushroom Monitoring with a Linear-Rail System.” GCREC Quarterly Tour, August 21, 2025.

Presenter. “AI-Based Mushroom Detection and Maturity Assessment.” National Clean Plant Network Annual Meeting tour, GCREC, April 16, 2025.

Presenter. “Linear-Rail Sensing for Mushroom Monitoring.” GCREC tour for White House staff, March 28, 2024.

Presenter. “Computer Vision for Mushroom Detection and Maturity Estimation.” GCREC tour for Senator Collins, February 24, 2023.

PEER REVIEW SERVICE

- Applied Engineering in Agriculture — 3 reviews
- IEEE Transactions on Computational Imaging — 2 reviews
- Journal of the ASABE — 2 reviews
- Smart Agricultural Technology — 5 reviews
- **Abstract Reviewer.** 2027 Florida Undergraduate Research Conference, November 2026.

ACADEMIC SERVICE

- **Session Moderator.** "AI, Machine Vision, and Intelligent Sensing for Agricultural Systems," ASABE Annual International Meeting, 2026.

LEADERSHIP EXPERIENCE

- **Communication Chair.** Gulf Coast Postdoc and Student Association (GCPSA), UF/IFAS Gulf Coast Research and Education Center | Jan-Dec 2025.

TECHNICAL SKILLS

Programming: Python, Julia, R, SQL, C, C++, Java | Computer vision and ML: RGB-D imaging, instance segmentation, object tracking, CNNs, LSTMs, YOLO, SVM | Sensing and hardware: ZED depth cameras, LiDAR, UAV hyperspectral imagery, Arduino, linear rail and stepper motor systems | Tools: Visual Studio Code, Jupyter Notebook, Roboflow, Qt.

ACADEMIC PROJECTS

AGRICULTURAL AND REMOTE SENSING PROJECTS

AI-Assisted Strawberry Genotype Calling from HRM Melt Curves *[Python]* 2026

- Developed LSTM-based genotype calling for six strawberry HRM markers.
- Developed a user interface that streamlines review of every well, saving time on manual checks.

Automated Detection and Mapping of White Button Mushrooms for Enhanced Farming Practices *[Python]* 2024

- Designed computer vision system to detect individual mushrooms and map their locations within growing beds for crop monitoring.

Advancing Mushroom Farming: Computer Vision for Detection and Maturity Estimation *[Python]* 2024

- Developed mushroom detection and segmentation methods alongside maturity assessment for selective harvesting research.

Land-Cover Classification with Hyperspectral and LiDAR Data Fusion *[Python]* 2022

- Combined hyperspectral pixels with local LiDAR neighborhoods and trained convolutional neural networks for land-cover classification.

Land-Cover Classification Using LiDAR Imagery *[Python]* 2022

- Converted LiDAR .laz file data from NSF NEON to a point cloud and visualized height and intensity of land cover.
- The statistics (3D height and intensity distribution) of each land cover class were computed to observe the difference between them.
- Performed classification with KNN using both single pixels and a 5x5 neighborhood as features. Experimented with KNN, SVM, and Random Forest for classification using the MUUFL Gulfport dataset and NEON NSF LiDAR data.

ADDITIONAL COMPUTER SCIENCE PROJECTS

Music Genre Classification *[Python]* 2022

- Extracted different features like chromagrams, spectrograms, Mel-spectrograms, and MFCC from music samples using the librosa library in Python.
- Developed a CNN, LSTM, and an ensemble of CNN and LSTM to classify music into different genres.
- Experimented with single features and combinations of different features to observe the performance of the models. The ensemble models with multiple features significantly outperformed the models trained on unimodal features.
- Additionally, conducted experiments with CNN autoencoders to reduce the dimensionality of spectrograms and performed classification using the latent space.

Face Recognition Using Contrastive Learning *[Python]* 2022

- A triplet network was used for classification since the number of samples was very few and imbalanced in the dataset.
- Developed a Triplet convolutional network to learn distinguishable embeddings of each class. Used KNN to make classification of test samples.
- Experimented with different numbers of triplets and different values of the margin parameter and observed the effect on performance.

Classification and Clustering Method Comparison *[Julia]* 2021

- Implemented Naïve Bayes, Decision Tree, Support Vector Machine, and Neural Network to perform classification. The performance of different methods was compared.
- For clustering, experimented with K-means, k-medoids, Fuzzy c-means, and Gaussian Mixtures Model. All these models were trained on four datasets: Car Evaluation, Abalone, Madelon, and KDD Cup 1999 dataset.

MIPS Simulator *[Python]* 2021

- Designed an instruction-by-instruction MIPS simulator to report register and data-memory contents during execution.

Language Translator *[Java, HTML, CSS, JavaScript]* 2021

- Designed a campus communication interface for students who do not speak English, with attention to usability and consistency.

Banking System Using Object-Oriented Programming *[C++]* 2019

- Created an account-management system with loan, foreign exchange, online banking, and card functions.

Online Customer Sales Management System *[SQL, Java]* 2019

- Designed a sales database and reporting interface for customers, products, branch sales, and profit summaries.

EARLIER INTERNSHIPS

Image Processing Intern, Instruments Research and Development Establishment, India (2020): Implemented face detection and feature-based tracking methods.

- The aim of this project was Object Detection and Tracking using Image Processing.
- Haar Cascade Classifier was used to detect face/ multiple faces from photos.
- Useful features for object tracking were also computed through the Harris Corner Detector and Shi-Tomasi algorithm. Performance of both algorithms was compared.
- Face tracking was done using features from the Shi-Tomasi algorithm.

Software Development Intern, Indian Institute of Technology Roorkee, India (2019): Worked on particle swarm optimization and differential evolution for engineering problems.

- Implemented the particle swarm optimization algorithm.
- Coordinated the development of variants of DE to solve discrete and combinatorial optimization problems.
- The proposed versions of DE were tested on Test examples comprising the Knapsack and Travelling Salesman Problems, and results were compared with the existing results.